

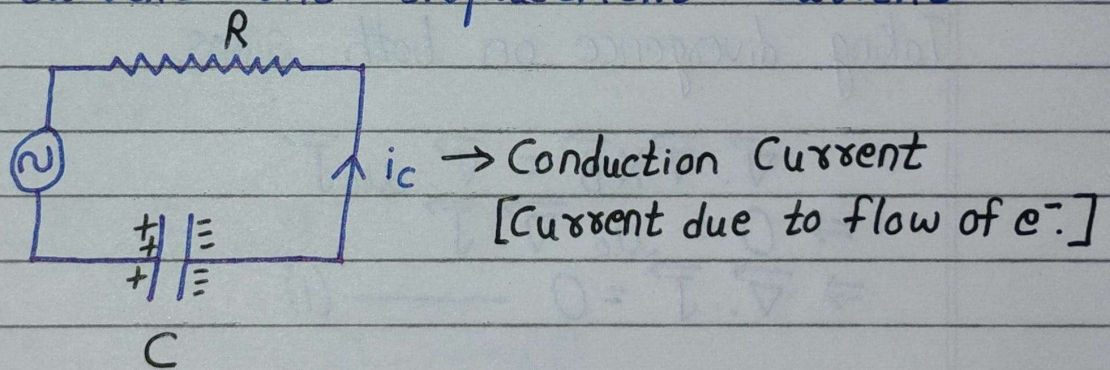
MODULE - III

Electromagnetic Theory

Displacement Current

There is no flow of electrons inside the dielectric between the plates of capacitor. To remove the discontinuity in the flow of current, J. C. Maxwell introduced the concept of displacement current.

“Displacement Current is an apparent current that flows between the plates of capacitor. It represents the rate at which change in charge takes place.” It flows only when the voltage is varying. Total current is equal to sum of conduction current and displacement current.



$$i_d = \frac{\partial q}{\partial t}$$

Inside the dielectric,
 $\vec{D} = \frac{q}{A}$

$$i_d = \frac{\partial \vec{D} A}{\partial t}$$

$$i_d = A \frac{\partial \vec{D}}{\partial t}$$

Displacement current density,

$$\vec{J}_d = \frac{i_d}{A} = \frac{\partial \vec{D}}{\partial t}$$

Total Current, $I = i_c + i_d$.

Imp # Modification of Ampere's Law

$$\vec{\nabla} \times \vec{B} = \mu_0 \vec{J}$$

Taking divergence on both sides,

$$\begin{aligned} \vec{\nabla} \cdot \vec{\nabla} \times \vec{B} &= \vec{\nabla} \cdot \mu_0 \vec{J} \\ \Rightarrow 0 &= \mu_0 \vec{\nabla} \cdot \vec{J} \\ \Rightarrow \vec{\nabla} \cdot \vec{J} &= 0 \quad \text{--- (1)} \end{aligned}$$

It means that, current is always closed, there is no source or sink. It is contradictory with Eqⁿ. of continuity given by eqⁿ. (2).

$$\vec{\nabla} \cdot \vec{J} + \frac{\partial \rho}{\partial t} = 0 \quad \text{--- (2)}$$

Maxwell concluded that eqⁿ. (1) is true only for static field. It must be modified for varying fields.

Acc. to Gauss's law in diff^e. form,

$$\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0}$$

$$\vec{\nabla} \cdot \vec{D} = \rho$$

Diff. w.r.to, t ,

$$\vec{\nabla} \cdot \frac{\partial \vec{D}}{\partial t} = \frac{\partial \rho}{\partial t}$$

Adding $\vec{\nabla} \cdot \vec{J}$ on both sides,

$$\vec{\nabla} \cdot \vec{J} + \vec{\nabla} \cdot \frac{\partial \vec{D}}{\partial t} = \vec{\nabla} \cdot \vec{J} + \frac{\partial \rho}{\partial t}$$

From eqⁿ. (2), RHS = 0

$$\vec{\nabla} \cdot \vec{J} + \vec{\nabla} \cdot \frac{\partial \vec{D}}{\partial t} = 0$$

$$\Rightarrow \vec{\nabla} \cdot \left(\vec{J} + \frac{\partial \vec{D}}{\partial t} \right) = 0 \quad \text{--- (3)}$$

Hence, total current must be sum of i_c & i_d .

$\therefore$ Modified Ampere's Law is,

$$\boxed{\vec{\nabla} \times \vec{B} = \mu_0 \left(\vec{J} + \frac{\partial \vec{D}}{\partial t} \right)}$$

Imp # Maxwell's Equations

1. Maxwell's First Equation $[\vec{\nabla} \cdot \vec{E} = \rho/\epsilon_0]$

Acc. to Gauss's law of electrostatics,

$$\int_S \vec{E} \cdot d\vec{S} = \frac{q}{\epsilon_0}$$

$$\Rightarrow \int_S \vec{E} \cdot d\vec{S} = \frac{1}{\epsilon_0} \int_V \rho \, dV \quad \left[\because \rho = \frac{q}{\Delta V} \right]$$

Using Gauss's divergence theorem,

$$\int_V \vec{\nabla} \cdot \vec{E} \, dV = \frac{1}{\epsilon_0} \int_V \rho \, dV$$

$$\Rightarrow \int_V \left(\vec{\nabla} \cdot \vec{E} - \frac{\rho}{\epsilon_0} \right) dV = 0$$

$$\Rightarrow \vec{\nabla} \cdot \vec{E} - \frac{\rho}{\epsilon_0} = 0$$

$$\Rightarrow \boxed{\vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0}}$$

$$\text{or, } \vec{\nabla} \cdot \epsilon_0 \vec{E} = \rho$$

$$\Rightarrow \boxed{\vec{\nabla} \cdot \vec{D} = \rho}$$

Integral Form:

Integrating first eqⁿ. of ~~ex~~ over arbitrary volume V enclosed by surface area S .

$$\int \vec{\nabla} \cdot \vec{E} dV = \int \frac{\rho}{\epsilon_0} dV$$

Using Gauss's divergence theorem,

$$\boxed{\int \vec{E} \cdot d\vec{S} = \frac{1}{\epsilon_0} \int \rho dV}$$

Physical Significance:

The net outward flux of electric field E through any closed surface enclosing a volume V is equal to the total charge contained within the volume divided ϵ_0 .

2. Maxwell's Second Equation $[\nabla \cdot \vec{B} = 0]$

Acc. to Gauss's law of magnetostatics,

$$\int_S \vec{B} \cdot d\vec{S} = 0$$

Using Gauss's divergence theorem,

$$\int_V \nabla \cdot \vec{B} dV = 0$$

$$\Rightarrow \boxed{\nabla \cdot \vec{B} = 0}$$

$$\Rightarrow \boxed{\nabla \cdot \vec{H} = 0} \quad [\because B = \mu_0 H]$$

Integral Form:

Integrating second eqⁿ. over an arbitrary volume V enclosed by surface area S .

$$\int_V \nabla \cdot \vec{B} dV = 0$$

Using Gauss's divergence theorem,

$$\boxed{\int_S \vec{B} \cdot d\vec{S} = 0}$$

Physical Significance:

The net outward flux of magnetic induction B through any closed surface is zero.

3. Maxwell's Third Equation $[\nabla \times \vec{E} = -\partial \vec{B} / \partial t]$

Acc. to Faraday's law of electromagnetic induction,

$$\oint_C E \, dl = -\frac{\partial \phi}{\partial t}$$

$$\Rightarrow \oint_C \vec{E} \, dl = -\frac{\partial}{\partial t} \left[\int_S \vec{B} \, dS \right] \quad \left[\because \phi = BA = \int \vec{B} \, dS \right]$$

$$\Rightarrow \oint_C \vec{E} \, dl = - \int_S \left[\frac{\partial \vec{B}}{\partial t} \right] dS$$

Using Stoke's theorem,

$$\int_S \nabla \times \vec{E} \, dS = - \int_S \left[\frac{\partial \vec{B}}{\partial t} \right] dS$$

$$\Rightarrow \int_S \left[\nabla \times \vec{E} + \frac{\partial \vec{B}}{\partial t} \right] dS = 0$$

$$\Rightarrow \nabla \times \vec{E} + \frac{\partial \vec{B}}{\partial t} = 0$$

$$\Rightarrow \boxed{\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}}$$

Integral Form:

Integrating third eqⁿ. over an arbitrary curve C enclosing surface S .

$$\int_S \vec{\nabla} \times \vec{E} \, dS = - \int \frac{\partial B}{\partial t} \, dS$$

Using Stoke's theorem,

$$\boxed{\int \vec{E} \, dl = - \frac{\partial}{\partial t} \left[\int_S \vec{B} \, dS \right]}$$

Physical Significance:

The E.M.F. induced around a closed path is equal to the negative rate of change of magnetic flux linked with a curve.

4. Maxwell's Fourth Equation $\left[\nabla \times \vec{B} = \mu_0 \left(\vec{J} + \frac{\partial \vec{D}}{\partial t} \right) \right]$

Acc. to Ampere's law in modified form,

$$\int \vec{B} d\ell = \mu_0 I = \mu_0 (i_c + i_d)$$

$$\therefore i_c = \int \vec{J} dS, \quad i_d = \int \vec{J}_d dS$$

$$\int \vec{B} d\ell = \mu_0 \left[\int \vec{J} dS + \int \vec{J}_d dS \right]$$

$$\Rightarrow \int \vec{B} d\ell = \mu_0 \int (\vec{J} + \vec{J}_d) dS$$

Using Stoke's theorem,

$$\int \nabla \times \vec{B} dS = \mu_0 \int (\vec{J} + \vec{J}_d) dS$$

$$\Rightarrow \int [\nabla \times \vec{B} - \mu_0 (\vec{J} + \vec{J}_d)] dS = 0$$

$$\Rightarrow \nabla \times \vec{B} = \mu_0 (\vec{J} + \vec{J}_d)$$

$$\Rightarrow \boxed{\nabla \times \vec{B} = \mu_0 \left(\vec{J} + \frac{\partial \vec{D}}{\partial t} \right)}$$

Integral Form:

Integrating fourth eqⁿ. over an arbitrary curve C enclosing surface S .

$$\int_S \vec{\nabla} \times \vec{B} dS = \int \mu_0 \left(\vec{J} + \frac{\partial \vec{D}}{\partial t} \right) dS$$

Using Stoke's theorem,

$$\boxed{\int \vec{B} d\ell = \mu_0 \int \left(\vec{J} + \frac{\partial \vec{D}}{\partial t} \right) dS}$$

Physical Significance:

The magnetomotive force induced around a closed path is equal to μ_0 times of total current.

Equation of Continuity

This eqⁿ. represents conservation of charges i.e. electrical charges can neither be created nor be destroyed.

$$\boxed{\vec{\nabla} \cdot \vec{J} + \frac{\partial \rho}{\partial t} = 0}$$

Proof:

Using Maxwell's Fourth Eqⁿ.,

$$\vec{\nabla} \times \vec{B} = \mu_0 \left(\vec{J} + \frac{\partial \vec{D}}{\partial t} \right)$$

Taking divergence on both sides,

$$\vec{\nabla} \cdot \vec{\nabla} \times \vec{B} = \vec{\nabla} \cdot \mu_0 \left(\vec{J} + \frac{\partial \vec{D}}{\partial t} \right)$$

$$\Rightarrow 0 = \mu_0 \left[\vec{\nabla} \cdot \vec{J} + \frac{\partial}{\partial t} (\vec{\nabla} \cdot \vec{D}) \right]$$

Using Maxwell's First Eqⁿ., $\vec{\nabla} \cdot \vec{D} = \rho$

$$\boxed{\vec{\nabla} \cdot \vec{J} + \frac{\partial \rho}{\partial t} = 0}$$

Electromagnetic Wave Propagation in Free Space

In free space, $\rho = 0$, $J = 0$, $\sigma = 0$

- Maxwell's eq^{ns} in free space,

$$(i) \quad \vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0}$$

$$\Rightarrow \vec{\nabla} \cdot \vec{E} = 0 \quad \text{--- (1)}$$

$$(ii) \quad \vec{\nabla} \cdot \vec{B} = 0 \quad \text{--- (2)}$$

$$(iii) \quad \vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \quad \text{--- (3)}$$

$$(iv) \quad \vec{\nabla} \times \vec{B} = \mu_0 \left(\vec{J} + \frac{\partial \vec{D}}{\partial t} \right)$$

$$\Rightarrow \vec{\nabla} \times \vec{B} = \mu_0 \left(0 + \frac{\partial}{\partial t} \epsilon_0 \vec{E} \right)$$

$$\Rightarrow \vec{\nabla} \times \vec{B} = \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t} \quad \text{--- (4)}$$

Eq^{ns}. (1), (2), (3) & (4) are Maxwell's eq^{ns}. in free space.

- Taking curl of eqⁿ. (3),

$$\vec{\nabla} \times \vec{\nabla} \times \vec{E} = -\vec{\nabla} \times \frac{\partial \vec{B}}{\partial t}$$

$$[\because \vec{a} \times \vec{b} \times \vec{c} = \vec{b}(\vec{a} \cdot \vec{c}) - (\vec{a} \cdot \vec{b})\vec{c}]$$

$$\vec{\nabla}(\vec{\nabla} \cdot \vec{E}) - (\vec{\nabla} \cdot \vec{\nabla}) \vec{E} = -\frac{\partial}{\partial t} (\vec{\nabla} \times \vec{B})$$

Using eq^{ns} (1) & (4),

$$0 - \nabla^2 \vec{E} = -\frac{\partial}{\partial t} \left(\mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t} \right)$$

$$\Rightarrow \boxed{\nabla^2 \vec{E} = \mu_0 \epsilon_0 \frac{\partial^2 \vec{E}}{\partial t^2}} \quad \text{————— (5)}$$

$$\Rightarrow \boxed{\nabla^2 \vec{D} = \mu_0 \epsilon_0 \frac{\partial^2 \vec{D}}{\partial t^2}} \quad [\because \vec{E} = \vec{D}/\epsilon_0]$$

Eqⁿ. (5) represents change in electric field w.r. to position and time in free space.

- Taking curl of eqⁿ. (4),

$$\begin{aligned} \vec{\nabla} \times \vec{\nabla} \times \vec{B} &= \vec{\nabla} \times \left(\mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t} \right) \\ \Rightarrow (\vec{\nabla} \cdot \vec{B}) \vec{\nabla} - (\vec{\nabla} \cdot \vec{\nabla}) \vec{B} &= \mu_0 \epsilon_0 \frac{\partial}{\partial t} (\vec{\nabla} \times \vec{E}) \end{aligned}$$

Using eq^{ns}. (2) & (3),

$$0 - \nabla^2 \vec{B} = -\mu_0 \epsilon_0 \frac{\partial}{\partial t} \left(\frac{\partial \vec{B}}{\partial t} \right)$$

$$\Rightarrow \boxed{\nabla^2 \vec{B} = \mu_0 \epsilon_0 \frac{\partial^2 \vec{B}}{\partial t^2}} \quad \text{————— (6)}$$

$$\Rightarrow \boxed{\nabla^2 \vec{H} = \mu_0 \epsilon_0 \frac{\partial^2 \vec{H}}{\partial t^2}}$$

$$[\because \vec{B} = \mu_0 \vec{H}]$$

Eqⁿ. (6) represents change in magnetic field w.r.to position and time in free space.

- The classical eqⁿ. of wave motion is,

$$\nabla^2 A = \frac{1}{v^2} \frac{\partial^2 A}{\partial t^2} \quad \text{--- (7)}$$

Comparing eq^{ns}. (7) with (5) or (6),

$$\frac{1}{v^2} = \mu_0 \epsilon_0$$

$$\Rightarrow v = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = \frac{1}{\sqrt{4\pi \times 10^{-7} \times 8.85 \times 10^{-12}}} = 3 \times 10^8 \text{ ms}^{-1}$$

$$\Rightarrow \boxed{c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}}$$

Transverse Nature of EM-wave or Solution of EM Wave Equation in Free Space

EM wave eq^{ns}. in free space are,

$$\nabla^2 \vec{E} = \mu_0 \epsilon_0 \frac{\partial^2 \vec{E}}{\partial t^2} \text{ --- (1)}$$

$$\nabla^2 \vec{B} = \mu_0 \epsilon_0 \frac{\partial^2 \vec{B}}{\partial t^2} \text{ --- (2)}$$

The solution of eqⁿ. (1) is,

$$\vec{E} = \vec{E}_0 \exp i(\vec{k} \cdot \vec{r} - \omega t) \text{ --- (3)}$$

$$\begin{aligned} \text{where, } \vec{k} \cdot \vec{r} &= (\hat{i}k_x + \hat{j}k_y + \hat{k}k_z) \cdot (\hat{i}x + \hat{j}y + \hat{k}z) \\ &= xk_x + yk_y + zk_z \end{aligned}$$

Maxwell's First Eqⁿ. in free space is,

$$\vec{\nabla} \cdot \vec{E} = 0$$

$$\Rightarrow \vec{\nabla} \cdot \vec{E}_0 \exp i(\vec{k} \cdot \vec{r} - \omega t) = 0$$

$$\begin{aligned} \Rightarrow \left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) \cdot (\hat{i}E_{0x} + \hat{j}E_{0y} + \hat{k}E_{0z}) \dots \\ \dots \exp i(xk_x + yk_y + zk_z - \omega t) = 0 \end{aligned}$$

$$\Rightarrow \frac{\partial}{\partial x} \left[E_{0x} \exp i(xk_x + yk_y + zk_z - \omega t) \right] \\ + \frac{\partial}{\partial y} \left[E_{0y} \exp i(xk_x + yk_y + zk_z - \omega t) \right] \\ + \frac{\partial}{\partial z} \left[E_{0z} \exp i(xk_x + yk_y + zk_z - \omega t) \right] = 0$$

$$\Rightarrow [i E_{0x} k_x \exp i(xk_x + yk_y + zk_z - \omega t)] \\ + [i E_{0y} k_y \exp i(xk_x + yk_y + zk_z - \omega t)] \\ + [i E_{0z} k_z \exp i(xk_x + yk_y + zk_z - \omega t)] = 0$$

$$\Rightarrow i[k_x E_{0x} + k_y E_{0y} + k_z E_{0z}] \exp i(\vec{k} \cdot \vec{r} - \omega t) = 0$$

$$\Rightarrow i \vec{k} \cdot \vec{E}_0 \exp i(\vec{k} \cdot \vec{r} - \omega t) = 0$$

$$\Rightarrow i \vec{k} \cdot \vec{E} = 0$$

$\Rightarrow \vec{E}$ is $\perp$ to direction of wave.

Similarly using $\vec{\nabla} \cdot \vec{B} = 0$,

$$i \vec{k} \cdot \vec{B} = 0$$

$\Rightarrow \vec{B}$ is $\perp$ to direction of wave.

Using Maxwell's Third Eqⁿ,

$$\vec{k} \times \vec{E} = \omega \vec{B}$$

$\Rightarrow \vec{B}$ is $\perp$ to $\vec{k}$ & $\vec{E}$ both.

$\Rightarrow$ EM wave is Transverse.

Poynting Vector

The rate at which energy is transmitted through unit area perpendicular to the direction of energy is called poynting vector $\vec{S}$.

$$\vec{S} = \vec{E} \times \vec{H}$$

It is also called Flux Vector and its unit is $\text{Joule (sec)}^{-1} (\text{m})^{-2}$.

The direction of $\vec{S}$ is parallel to the direction of propagation of energy.

~~V. Imp.~~ # Poynting Theorem

Maxwell's Third & Fourth eq^{ns}. are,

$$\vec{\nabla} \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \quad \text{———— (1)}$$

$$\vec{\nabla} \times \vec{B} = \mu_0 \left(\vec{J} + \frac{\partial \vec{D}}{\partial t} \right)$$

$$\Rightarrow \vec{\nabla} \times (\mu_0 \vec{H}) = \mu_0 \left(\vec{J} + \frac{\partial \vec{D}}{\partial t} \right)$$

$$\Rightarrow \vec{\nabla} \times \vec{H} = \vec{J} + \frac{\partial \vec{D}}{\partial t} \quad \text{———— (2)}$$

$$\therefore \vec{\nabla} \cdot (\vec{E} \times \vec{H}) = \vec{H} \cdot (\vec{\nabla} \times \vec{E}) - \vec{E} \cdot (\vec{\nabla} \times \vec{H})$$

$$\vec{\nabla} \cdot (\vec{E} \times \vec{H}) = -\vec{H} \cdot \frac{\partial \vec{B}}{\partial t} - \vec{E} \cdot \left(\vec{J} + \frac{\partial \vec{D}}{\partial t} \right)$$

$$\Rightarrow \vec{\nabla} \cdot (\vec{E} \times \vec{H}) = -\vec{H} \cdot \frac{\partial (\mu \vec{H})}{\partial t} - \vec{E} \cdot \vec{J} - \vec{E} \cdot \frac{\partial (\epsilon \vec{E})}{\partial t}$$

$$\because \frac{\partial H^2}{\partial t} = 2\vec{H} \cdot \frac{\partial \vec{H}}{\partial t} \quad \& \quad \frac{\partial E^2}{\partial t} = 2\vec{E} \cdot \frac{\partial \vec{E}}{\partial t}$$

$$\vec{\nabla} \cdot (\vec{E} \times \vec{H}) = -\mu \frac{1}{2} \frac{\partial H^2}{\partial t} - \epsilon \frac{1}{2} \frac{\partial E^2}{\partial t} - \vec{E} \cdot \vec{J}$$

$$\Rightarrow \vec{\nabla} \cdot (\vec{E} \times \vec{H}) = -\frac{\partial}{\partial t} \left[\frac{\mu H^2 + \epsilon E^2}{2} \right] - \vec{E} \cdot \vec{J}$$

Integrating over volume V ,

$$\int \vec{\nabla} \cdot (\vec{E} \times \vec{H}) dV = -\int \frac{\partial}{\partial t} \left[\frac{\mu H^2 + \epsilon E^2}{2} \right] dV - \int \vec{E} \cdot \vec{J} dV$$

Applying divergence theorem,

$$\boxed{\int_{\text{I}} \vec{E} \times \vec{H} dS = -\frac{\partial}{\partial t} \left[\int_{\text{II}} \left(\frac{\mu H^2 + \epsilon E^2}{2} \right) dV \right] - \int_{\text{III}} \vec{E} \cdot \vec{J} dV}$$

Physical Significance:

- III-term represents the rate of transfer of energy due motion of charges.

- II-term represents the rate of stored electromagnetic energy.
- I-term represents the total amount of energy crossing per second through closed surface.

Therefore, $\vec{E} \times \vec{H}$ represents the amount of energy transferred per unit time per unit surface area.

Q. Show that: $\frac{\vec{E}}{\vec{B}} = c$

From solving eqⁿ. Maxwell's Third Eqⁿ.,

$$\vec{k} \times \vec{E} = \omega \vec{B}$$

$$\therefore \vec{E} \text{ is } \perp \vec{k}$$

$$\Rightarrow kE = \omega B$$

$$\Rightarrow \frac{\vec{E}}{\vec{B}} = \frac{\omega}{k} = \frac{2\pi \nu \times \lambda}{2\pi} = c.$$

Q. Show that: $\frac{\vec{E}}{\vec{H}} = \sqrt{\frac{\mu_0}{\epsilon_0}}$

$$\frac{E}{B} = c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$$

$$\frac{E}{\mu_0 H} = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$$

$$\Rightarrow \frac{E}{H} = \sqrt{\frac{\mu_0}{\epsilon_0}}.$$

Numericals

Q. If Earth receives $2 \text{ cal min}^{-1} \text{ cm}^{-2}$ solar energy, what are the amplitude of electric and magnetic fields of radiation?

Solⁿ:

$$|\vec{S}| = |\vec{E} \times \vec{H}| = 2 \text{ cal min}^{-1} \text{ cm}^{-2}$$

$$\Rightarrow EH \sin \theta = \frac{2 \times 4.2}{60 \times 10^{-4}} \text{ J sec}^{-1} \text{ m}^{-2}$$

$$\Rightarrow EH = 1400 \text{ ——— (1)}$$

We have,

$$\frac{E}{H} = \sqrt{\frac{\mu_0}{\epsilon_0}} = 376.72 \text{ ——— (2)}$$

Multiplying eqⁿ. (1) by (2),

$$EH \cdot \frac{E}{H} = 1400 \times 376.72$$

$$\Rightarrow E = 726.22 \text{ V/m}$$

$$\text{Also, } H = \frac{1400}{726.22} = 1.927 \text{ A/m}$$

$$E_0 = E\sqrt{2} = 1027.030 \text{ V/m}$$

$$H_0 = H\sqrt{2} = 2.725 \text{ A/m}$$

Q. Assuming that all the energy from a 1000 Watt lamp is radiated uniformly, calculate the average values of intensities of an electric and magnetic field of radiation at a distance of 2m from the lamp?

Solⁿ:

$$\text{Power} = 1000 \text{ Watt} = 1000 \text{ Js}^{-1}$$

$$|\vec{S}| = |\vec{E} \times \vec{H}| = \frac{\text{Power}}{\text{Area}} = \frac{1000}{4\pi r^2}$$

$$\Rightarrow EH \sin\theta = \frac{1000}{4\pi (2)^2}$$

$$\Rightarrow EH = \frac{1000}{16\pi} \text{ ————— (1)}$$

$$\Rightarrow \frac{E}{H} = \sqrt{\frac{\mu_0}{\epsilon_0}} = 376.72 \text{ ————— (2)}$$

Multiplying eqⁿ. (1) by (2),

$$EH \cdot \frac{E}{H} = \frac{1000}{16\pi} \times 376.72$$

$$\therefore E = 86.57 \text{ V/m}$$

$$\text{Also, } H = \frac{1000}{16\pi \times 86.57} = 0.2298 \text{ A/m}$$